PDS CAPSTONE PROJECT REPORT

Introduction to Programming and Data Structures Lab (CS1P001)

LINE FOLLOWER

(WITH OBSTACLE DETECTION)

TEAM

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INTRODUCTION

Line Following is one of the most important aspects of robotics. A Line Following Robot is an autonomous robot which is able to follow either a black line that is drawn on the surface consisting of a contrasting color. It is designed to move automatically and follow the line. The robot uses arrays of optical sensors to identify the line, thus assisting the robot to stay on the track. The array of two sensor makes its movement precise and flexible. The robot is driven by DC gear motors to control the movement of the wheels. The Arduino Uno interface is used to perform and implement algorithms to control the speed of the motors, steering the robot to travel along the line smoothly. The ultrasonic sensor present in the robot detects any type of obstacle in front of the robot so that there is no accident happening with the robot. This project aims to implement the algorithm and control the movement of the robot by proper tuning of the control parameters and thus achieve better performance. It can be used industrial automated equipment carriers, small household applications, tour guides in museums and other similar applications, etc.

MOTIVATION

We went through many topics when we were given this project. But the line follower fascinated us the most as there are many practical applications for this project. Ever thought of a robot which follows line? A perfect or near perfect mimic of nature? After all, the purpose of robotics is to recreate in terms of machines what one see around to solve a problem or fulfil a requirement . Developing a line follower robot involves overcoming challenges and solving problems related to sensor calibration, line detection algorithms etc. We thought that this alone won't challenging enough for us. So, we decided to add obstacle avoidance part in this robot.

Our fascination towards robotics and desperation to learn more about building and programming robots is also one of the reasons for choosing this topic.

The areas that will be benefitted from the project:

- Industrial automated equipment carriers
- Entertainment and small household applications
- Tour guides in museums and other similar applications

HIGHLIGHTS OF THE PROJECT

1)The robot is capable of following a line using IR sensors.

2)It can be able to take various degrees of turns.

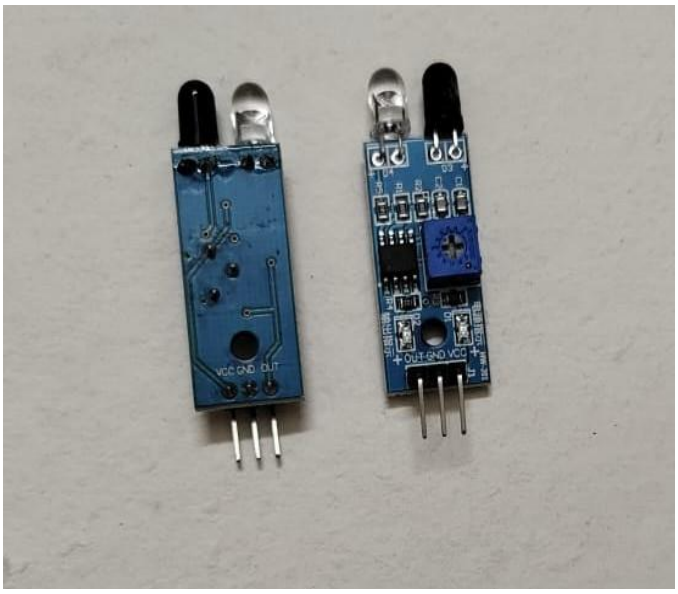
3)The robot is insensitive of environmental factors such as lighting and noise.

4)The robot is equipped with ultrasonic sensor which detects obstacles in the path and acts accordingly.

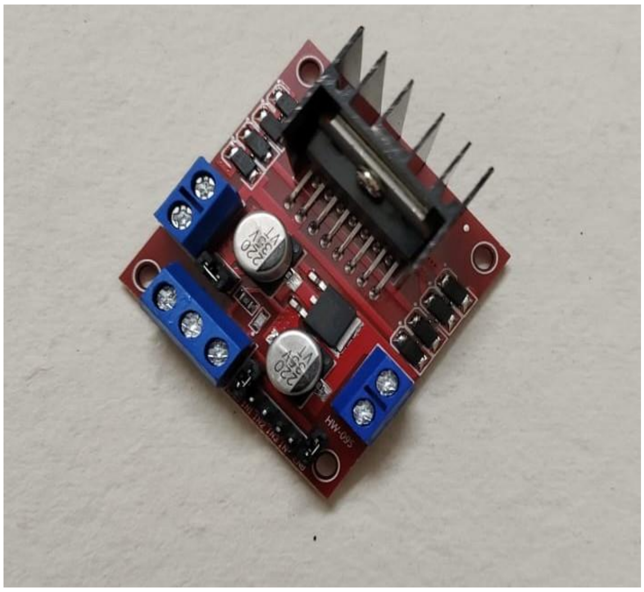
COMPONENTS

ULTRASONIC SENSOR

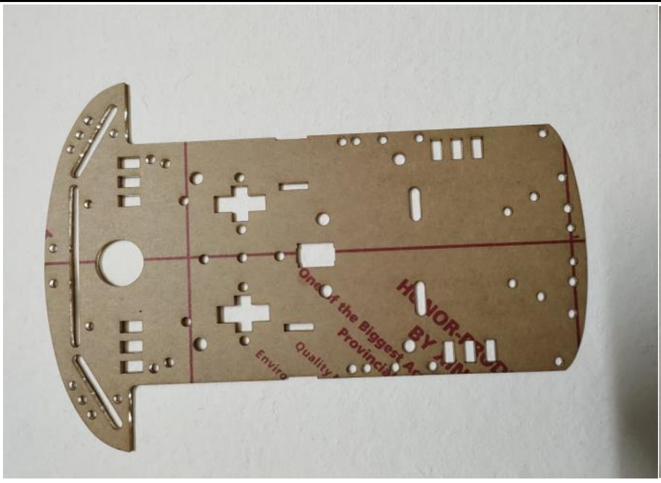
ARDUINO UNO AND USB CABLE



IR SENSORS

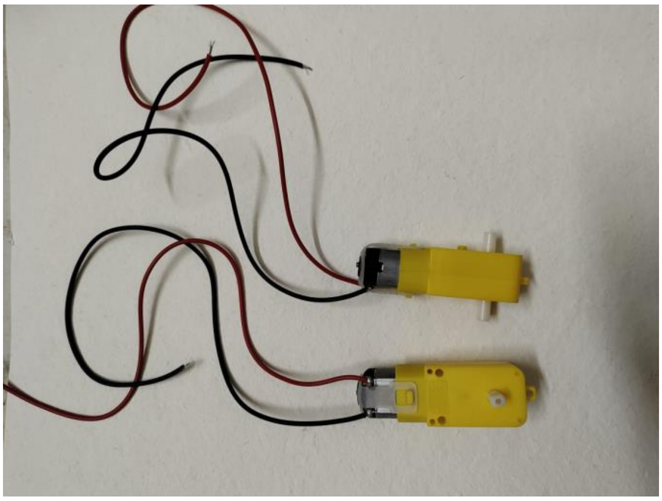


L2987N MOTOR DRIVER MODULE



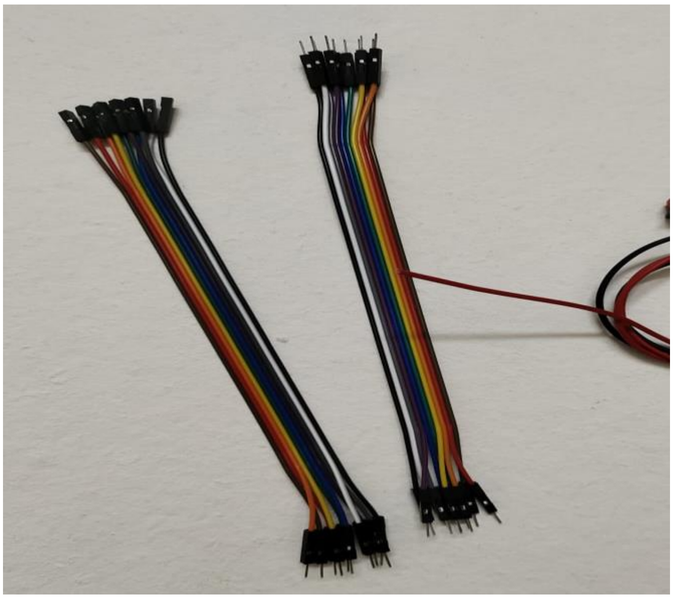
CAR CHASIS

CONNECTORS



WHEELS

GEARED MOTORS



JUMPER WIRES

BREAD BOARD

BATTERY HOLDER

BATTERIES

WHAT DID WE LEARN FROM THIS PROJECT?

1)Arduino UNO:

- The Arduino UNO is an open-source microcontroller board based on the Microchip ATmega328P microcontroller and developed by Arduino.cc. The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards (shields) and other circuits. The board has 14 digital I/O pins (six capable of PWM output), 6 analog I/O pins, and is programmable with the Arduino IDE (Integrated Development Environment), via a type B USB cable. It can be powered by a USB cable or a barrel connector that accepts voltages between 7 and 20 volts, such as a rectangular 9-volt battery.
- It is our first time of using Arduino UNO. We initially started learning basics like Blinking of LED lights etc.
- We came out of our comfort zone and went through Internet to learn about the syntax of Arduino and many new concepts.
- Resources: New Arduino Tutorials
  https://youtube.com/playlist?list=PLGs0VKk2DiYw-L-RibttcvK-WBZm8WLEP

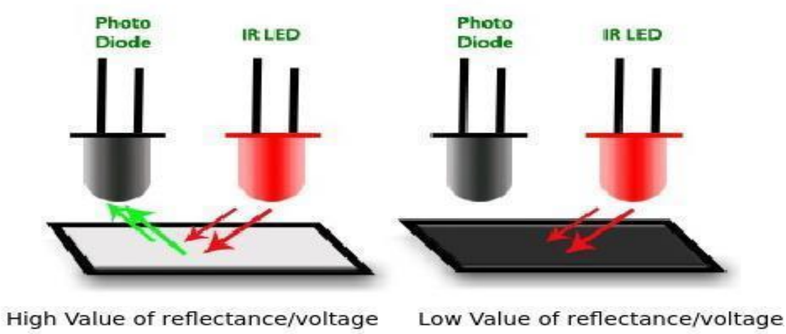
2)Ultrasonic library:

- This library is very handy. It greatly reduced the length of our code .
- The syntax for this library is such as
  variable = sonar.read();

- Here the variable stores the data of the distance of the object from the obstacle.
- The ultrasonic sensor sends echo waves and measures the distance of the obstacle from the robot, this process occurs continuously and the robot acts accordingly

3)IR Sensors:

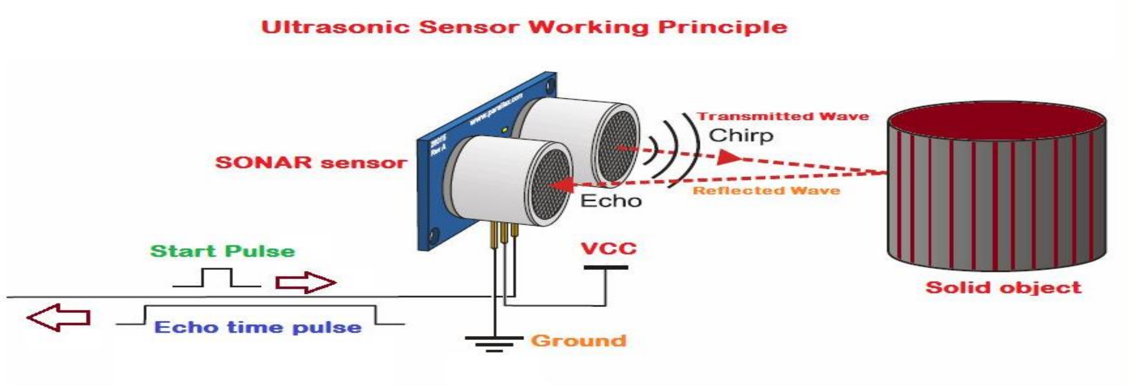
- An infrared (IR) sensor is an electronic device that measures and detects infrared radiation in its surrounding environment.
- Anything that emits heat (everything that has a temperature above around 5 Kelvin) gives off infrared radiation.
- Infrared sensors work with radar technology which emit and receive infrared radiation. This radiation hits the objects nearby and bounces back to the receiver of the device. Through this technology, the sensor can not only detect movement in an environment but also how far the object is from the device.



- In this project the radiation emitted by IR sensor is completely absorbed by the black line which helps to understand the path of the robot.

4)Ultrasonic sensor

- Ultrasonic sensors are based on the measured propagation time of the ultrasonic signal. They emit high-frequency sound waves which reflect on an object. The objects to be detected may be solid, liquid, granular or in powder form.

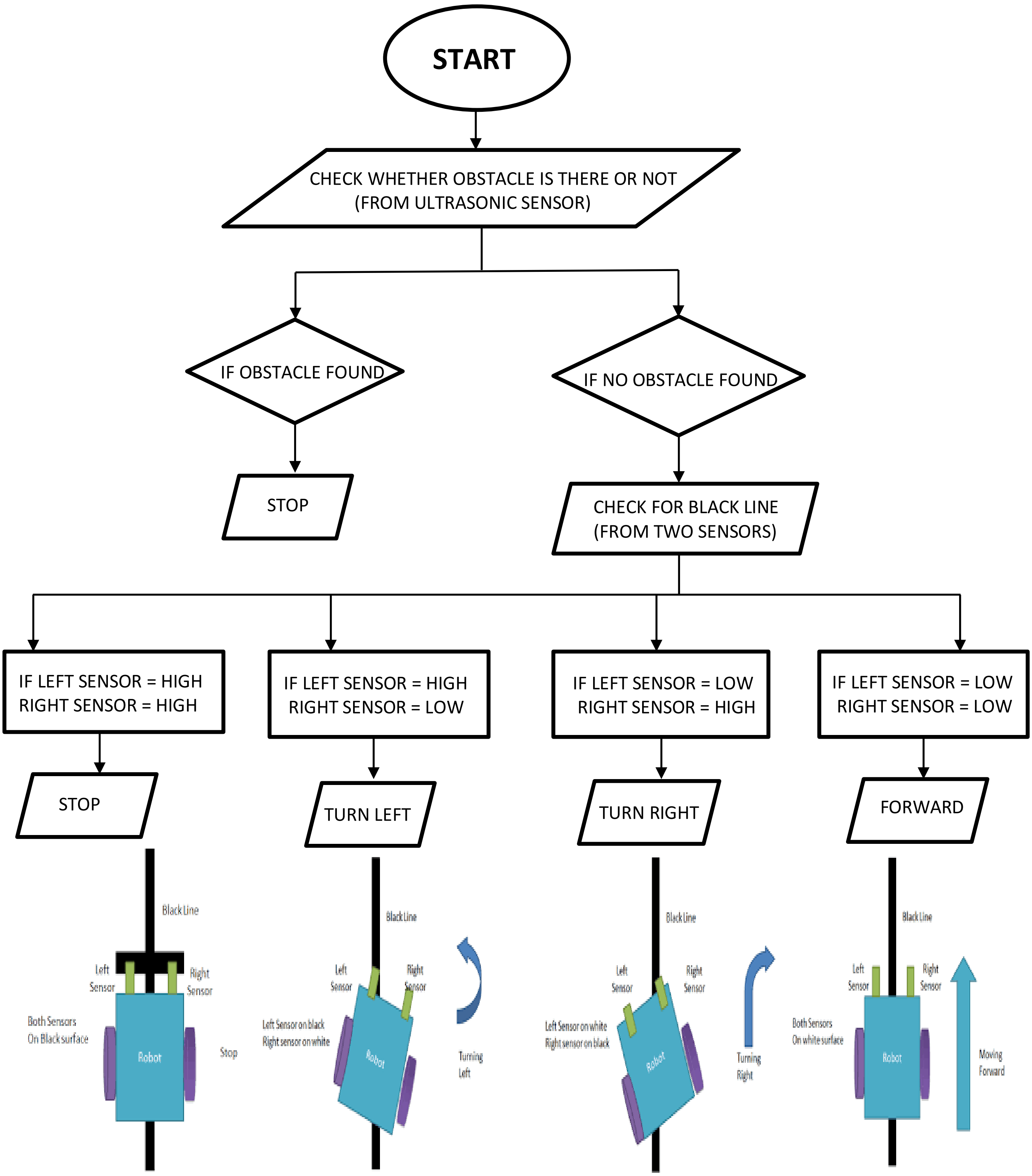


- This process helps to identify the object in the path of the robot.

5)Motor drive Module

- The L298N is a dual H-Bridge motor driver which allows speed and direction control of two DC motors at the same time. The module can drive DC motors that have voltages between 5 and 35V, with a peak current up to 2A.

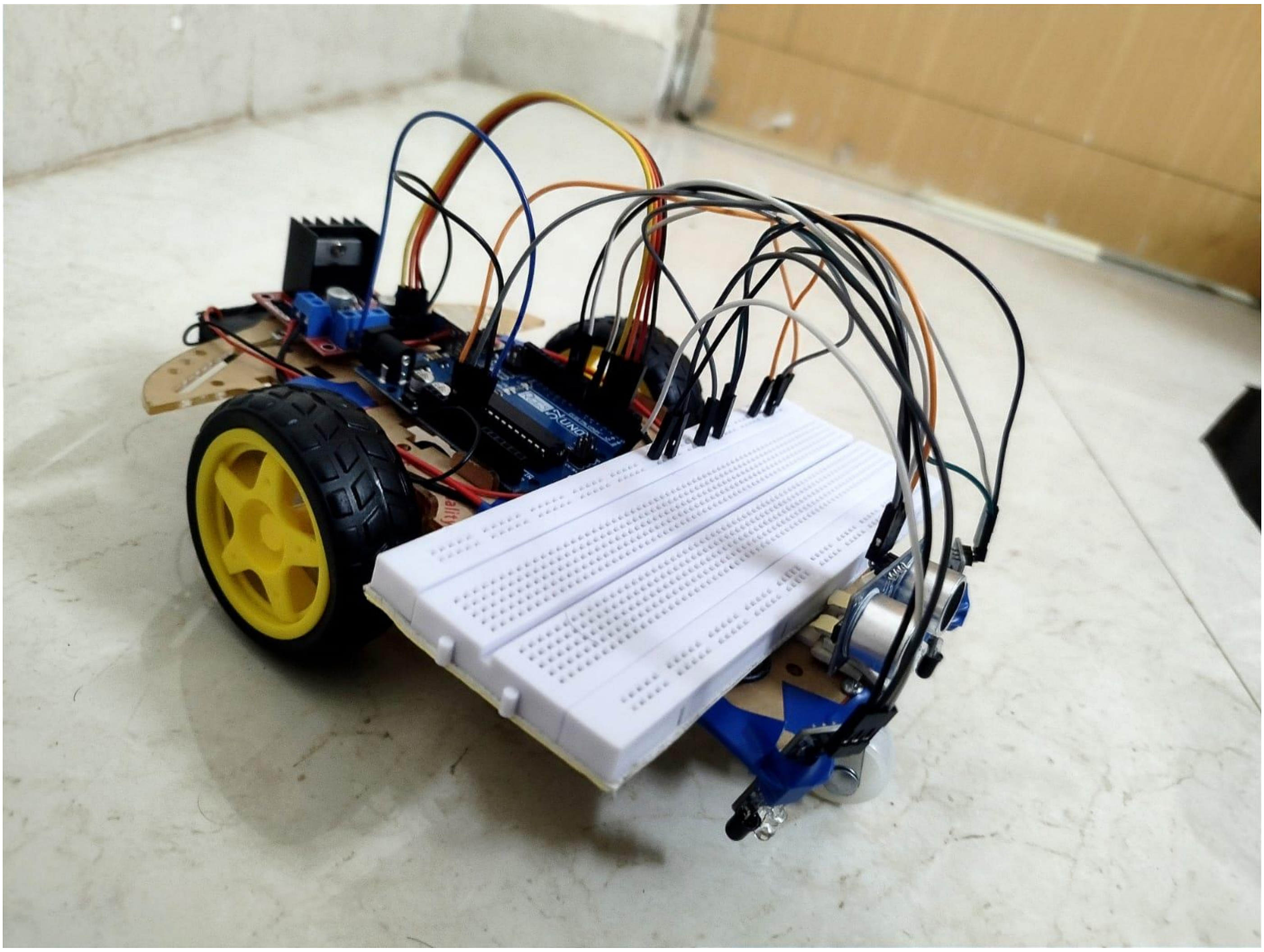
WORKING PRINCIPLE / ALGORITHM



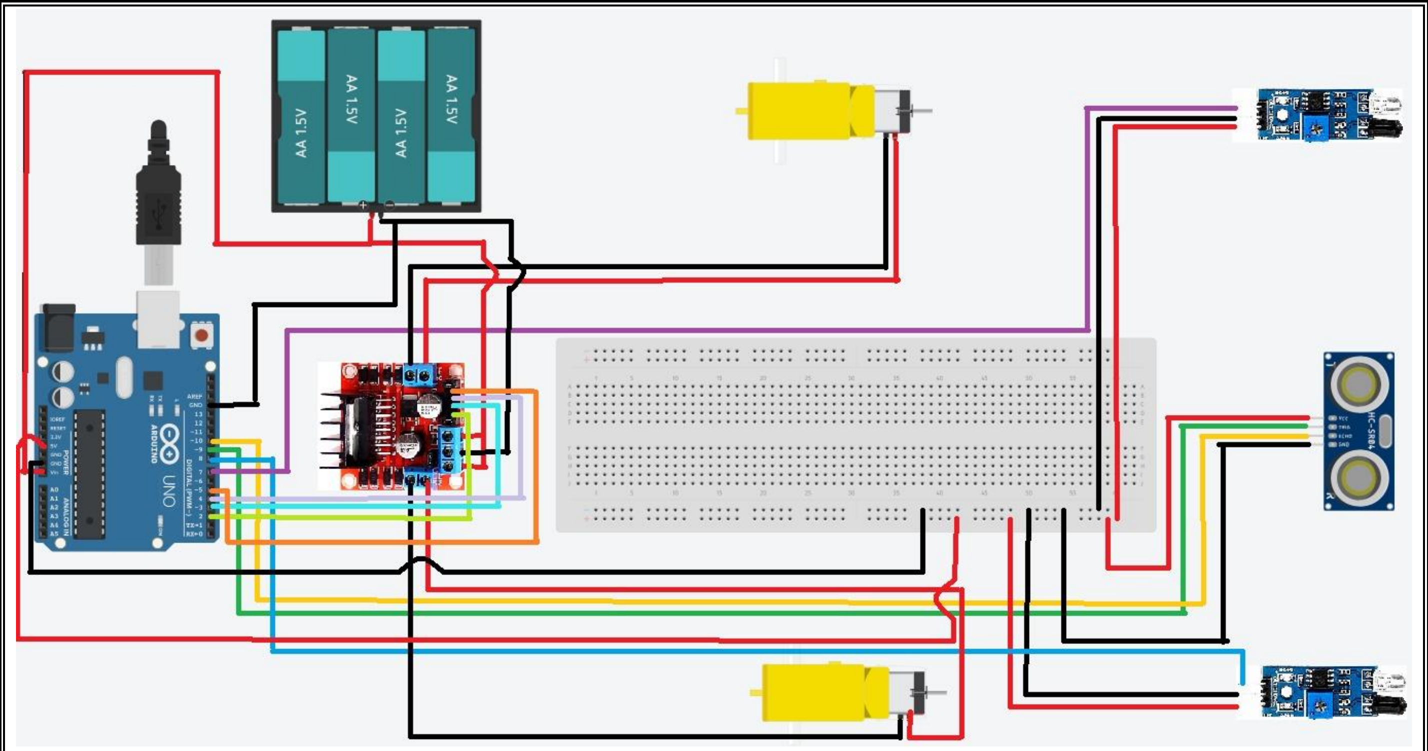
WORKING

The logic behind the working of Arduino is to analyse the input from the sensor according to program fed to it and provide corresponding output to the motor in such way that, it produce required motion.

THE LINE FOLLOWER ROBOT



CIRCUIT DIAGRAM



AREAS OF IMPROVEMENT

1)Colour sensor: We thought of adding a colour sensor to the robot so that it can recognize different colours and also follow the traffic rules. The colour sensor can be used to vary the speed of the robot by fixing different speed for different colours.

2)Installation of CCD camera: CCD camera can be used for better recognization and precise tracking the path.

3)LCD Screen: To display the total distance covered by the line follower.

4)Installation of servo motor: Servo motor can to be used to see the tracks in possible directions which makes it to move in the correct path and can even avoid obstacles and take different path.

FUTURE SCOPE

1)Autonomous Vehicles: Line following is a fundamental concept in autonomous vehicle navigation. Many autonomous vehicles, such as self-driving cars and drones, utilize similar techniques to follow lanes or paths.

2)It reduces traffic and drastically reduces road accidents because of ultra sonic sensors which automatically stops the automobile when any other vehicle comes near it.

3)Agricultural Automation: Line following technology can be applied to agricultural robots or drones for tasks like crop monitoring, irrigation, and pesticide application. These robots can follow lines or rows in fields to perform targeted actions, optimizing agricultural practices, and reducing human effort.

4)Line follower projects serve as a foundation for more advanced research and prototyping in the field of robotics and automation. They can help us develop and test new algorithms, sensor technologies, and control systems for autonomous navigation.

CONTRIBUTIONS

1)Installation of hardware:
Mohith, Vijay, Harsha.

2)Circuit connections for line follower:

Vijay, Harshith, Rohith.

3)Updated Circuit connections of line follower for obstacle detection:
Mohith, Harsha, Rohith.

4)Code for line follower:
Vijay, Harshith, Rohith.

5)Updated Code of line follower for obstacle detection:
Everyone did their individual contribution.

6)Preparation of pdf:
Harsha, Mohith, Harshith.

LESSONS LEARNT

- From this project we learnt a lot about hardware equipment which is useful for our future.
- Even though conceptually it will seem simple, when you start doing it you may face a lot of problems.
- The code of this project looks simple but practical implementation of hardware is very problematic.
- The main thing we learned from this project is being patient even the equipment has burnt several times we keep on pushing ourself to make this Project complete success.